

Recommended Bulgarian TST Problems

To aid self-study, here I've arranged all the theoretical problems that I like, by topic. I've mostly selected for pedagogical value, but I've also attempted to strike a good balance between easier and harder problems. The format I use for this list is "year/exam type/problem number". For each problem, I've added an estimate for the difficulty, as follows:

(●) – easy (●) – moderate (●) – difficult (●) – very difficult

You might disagree, and that's normal! In case you're interested in a specific subject, I've also provided descriptions for what each problem is about. Inevitably, some hints will slip through, so if you prefer to [onsight](#) the problems rather than [flashing](#) them, consider this your warning. Note that I've deliberately avoided cataloguing the problems from 2014 and 2026 so that those can be done as mock exams.

Mechanics

Statics

2018 T 3 Car suspension (●). This is quite tough if you don't know how cars work, because you'll have to learn on the go. Otherwise the physics isn't too bad.

Not Rotation

2019 T 3 Hodograph (●). A rather pleasant problem which introduces a type of velocity diagram.

2016 T 2 Friction on a rotating disk (●). A problem with a neat final result that tests your approximation skills.

2008 T 3 Deflection (●). An important exercise which shows the utility of the centre of mass frame in collision problems.

Rotation

2019 T 1 Unwinding a string (●). A short exercise on no-slip rotation.

2008 S 1 Ball rolling down a ball (●). A generic rotation problem with friction and normal forces.

2012 S 1 Falling rod (●). A problem which requires you to spot a kinematic constraint. Inspect for geometrical curiosities and proceed from there.

2018 S 1 Falling ladder (●). Similar to the above, albeit a bit more difficult.

2016 T 3 Masses on a rotating spring (●). A neat one – combines a few distinct ideas.

2023 T 3 Pendulum on an incline (●). You should get some practice on 3D rotation (e.g. via Kevin Zhou M8) before attempting this one.

Oscillations

2012	T	2	A sheet of paper (●). Moments of inertia and a physical pendulum.
2016	T	1	Leaning rectangle (●). A somewhat tedious problem which requires you to be very careful. As usual with problems on complicated oscillations, it's better to work with energy rather than Newton's second law.
2018	T	1	Blocks on a spring (●). Solving this rigorously is a nightmare, but you can still obtain some answer which is likely to be right, and promptly give up on analysing the complete motion.
2023	S	1	Two collisions (●). As with the previous problem, you need to be comfortable with differential equations. There's also some neat maths waiting for you at the end.

Gravity

2008	T	1	Drag on a satellite (●). A fun little problem on work and energy.
2016	S	1	Earth and Mars (●). Some standard fare on Kepler's laws.

Fluids

2018	T	2	Spray (●). A useful exercise on ideal fluids and inertial forces.
2008	T	2	Poiseuille's law (●). A standard problem on viscous flow in a cylinder. This pops up often in Olympiads, so you should probably know how to derive this by heart.
2023	T	2	The Danube (●). Viscous flow on an incline.

Heat

Ideal gas processes

2018	S	3	Bose-Einstein condensation (●). A semi-classical treatment of an interesting quantum phenomenon. Not as scary as it looks.
2018	T	8	Fast and slow (●). A key problem on nonquasistatic processes, with some non-trivial maths at the end.

Heat engines

2023	T	10	Three processes (●). This isn't that difficult, but note that you must stay agnostic about the equation of state. Otherwise you'd get zero marks here.
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Radiation

2019	T	4	Wien's law (●). This guides you through the derivation of Wien's law from Planck's law. Mostly a maths problem.
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Kinetic theory

2018	T	9	Constant collision rate (●). A neat, but somewhat artificial problem on mean free path.
2019	T	10	Helium (●). Two problems on effusion and momentum transfer, respectively. You're not required to get the numerical prefactors exactly right.
2021	S	3	Tea in a vacuum flask (●). An open-ended problem which requires deep insight into both radiation and kinetic theory.

Electromagnetism

Electrostatics

2017	S	1	Charge in a dielectric medium (●). A not-too-involved problem on conservation laws.
2009	T	4	Dipoles (●). Derivations of some standard formulae for electric dipoles.
2018	T	4	Field strength (●). The first part of this is alright, but the second involves a standard trick which is nevertheless hard to figure out on your first encounter.
2019	S	2	Conducting sphere (●). A finicky image charge problem.

Ohm's law

2014	T	4	A microscopic model for resistance (●). A short problem on a variation of the Drude model.
2018	T	5	Accelerating ring (●). A one-off problem about the Tolman-Stewart effect.
2021	S	2	Discharge (●). Electrostatics and Ohm's law.
2013	T	4	Resistivity of water (●). A pedagogical problem on the superposition of current distributions.

Lorentz force

2009	S	2	Crossed fields (●). A classic problem where you need to take components.
2012	T	3	Crossed fields redux (●). This is identical, but with more maths. Still, it's good practice for solving differential equations.

Induction

2015	T	6	Charged disk in a magnetic field (●). A problem on Faraday's law in which you'll need to set up an integral for the net torque.
2023	S	2	Betatron (●). A devious problem on a relativistic particle accelerator.
2018	S	2	Induction motor (●). A very pretty problem on an industrial device. You might reach an intimidating set of equations, but do not despair, they can simplify.

Circuits

2009	T	5	Pyramid (●). A problem on equivalent resistance for which you'll need the usual tricks for simplifying nodes.
2007	T	4	IV curve (●). A standard problem on Kirchoff's laws. It also tests on error propagation formulae.
2007	T	5	LR circuit (●). You'll need to know how inductors work under a sudden change in the circuit, but otherwise it's smooth sailing.
2011	S	2	LCR circuit (●). This one seems quite tedious, but it is possible to avoid the maths if you can spot the trick.
2023	T	4	Voltage rectifier (●). An electronics problem with casework. It's possible to solve it analytically, but making the correct approximations requires strong physical intuition.

Optics and waves

Geometrical optics

2017	T	5	Two images (●). A school problem on the thin lens formula.
2019	T	5	Prism (●). A longer problem on reflection and Snell's law. Lots and lots of trigonometry.
2023	T	6	Optical fiber (●). An elegant problem on signal propagation.

Wave optics

2017	T	6	Coincident maxima (●). Mostly maths, but still somewhat entertaining.
2011	T	6	Rotating the grating (●). A problem on optical path differences which requires prudent approximation.
2009	T	6	Billet split lens (●). A classic setup for observing interference. However, I don't know of a simple solution to Part (c). My own attempt is basically a whole essay, and I still wouldn't call it rigorous!
2018	T	6	Lens and plate (●). A setup for interference which I haven't seen anywhere else. Think carefully about where the interfering rays might come from.
2007	T	6	The width of a maximum (●). A problem on the intensity distribution of an N -slit setup.
2023	T	5	Convolutd grating (●). A formidable problem which you can still handwave your way around. Try phasors rather than complex numbers.
2013	T	6	Prism monochromator (●). An absolute unit of a problem. For a warm-up, maybe attempt RMPH2017-1A first.

Waves

2019	S	1	Linear crystal (●). A neat introduction to the physics of phonons. I remember finding the problem statement a bit too vague.
2019	S	3	Surface gravity waves (●). A scary-looking variation of the standard problem on wave reflection and transmission.

Modern physics

Relativity

2010	T	3	Beta decay (●). An exercise on the conservation laws.
2023	T	7	Scattering (●). Some more relativistic dynamics, this time in 2D. Try not to give up on the algebra too early.
2018	T	7	Relativistic force (●). A problem on $F = dp/dt$ with lots of integration.
2019	T	9	Accelerator (●). A teeny-tiny problem on a particle accelerator. However, take caution as to how you interpret the problem statement.

Heisenberg's uncertainty principle

2018	T	10	Atomic beam (●). A troll problem. If you're not opposed to a major hint, check out IPhO2001 1D first.
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Semiclassical atoms and molecules

2023	T	8	Predictions of the Bohr model (●). A conceptually straightforward problem, but getting the numerical values right is quite the ordeal.
2019	T	8	Transition energy (●). A difficult-to-interpret problem on conservation laws and energy levels in hydrogen.
2023	S	3	Yukawa potential (●). A subtle problem on a semiclassical model of the deuteron. Try to adhere to HUP and energy minimisation, and you'll be alright.

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